RadarML: A Multi-Channel Over-the-Air Dataset of Common Radar Waveforms for RESM Evaluation

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***Abstract* — This paper introduces "RadarML", a publicly available dataset of modulated radar pulses recorded over-the-air across multiple semi-independent receiver channels. The dataset was generated using the UCL ARESTOR platform which is a Radio-Frequency-System-on-a-chip (RFSoC) based Software Defined Radio (SDR). The dataset contains a library of around two million waveforms with varying centre frequencies, bandwidths, pulse widths and modulations and 1,000 examples of each signal. The goal of this dataset is to enable comparative analysis of radar modulation classification techniques on real data and stimulate research into multi-channel signal detection and modulation classification methods.**

***Keywords* — Radar, ESM, ELINT, Software Defined Radio.**

I. Introduction

As utilisation of the Radio Frequency (RF) spectrum continues to grow, the density of emissions within any given frequency band increases. For Electronic Surveillance (ES) systems, the ability to reliably detect and classify radar signals has therefore become increasingly important. Modern radar systems, particularly software defined radars are waveform agile and capable of adapting parameters such as centre frequency, bandwidth, pulse duration, modulation, and Pulse Repetition Frequency (PRF) on a pulse-to-pulse basis. This agility, combined with the proliferation of complex modulation schemes, creates a significantly more challenging electromagnetic environment for ES systems.

There are many publications that deal with the problem of signal detection and modulation classification of radar signals. However, many of these papers use different definitions and implementations of pulse modulation types making comparative analysis of techniques difficult, and very few utilise data that was captured experimentally. As far as the authors of this paper are aware, none use multiple co-located receiver channels simultaneously capturing the same radar pulses.

To address this gap in available data, we present "RadarML", a publicly available dataset of both frequency and phase modulated radar pulses that were captured experimentally over-the-air across multiple Analogue-To-Digital (ADC) channels using a Radio-Frequency-System-on-a-Chip (RFSoC) based system.

The aim of "RadarML" is to enable comparative analysis of radar modulation classification and parameter extraction techniques by serving as a standard reference dataset and to stimulate research into new radar electronic surveillance techniques that use multiple semi-independent receive channels to enhance detection and modulation classification performance.

II. Dataset Description

The dataset contains a broad range of representative radar waveforms with parameters as described below in Table 1. These waveforms vary in centre frequency, time duration or bandwidth (if applicable). Many of the waveforms are derived from open-source literature [1], however some more exotic examples such as non-linear chirps are included, which are more challenging than traditional radar waveforms for ES techniques. For each permutation of waveform there was 1,000 pulses generated.

Table 1. Synthetic Waveform Library

Modulation	**Baseband Centre Frequencies (MHz)**	**Bandwidths (MHz)**	**Pulse Widths (μs)**
CW	-10, -30, 30, +20	N/A	10, 50, 100
Linear FM	-30, -10, 0, 10, 30	+/- 10, 20, 40, 80	10, 50, 100
Exponential FM	-30, -10, 0, 10, 30	+/- 10, 20, 40, 80	10, 50, 100
Quadratic FM	-30, -10, 0, 10, 30	+/- 10, 20, 40, 80	10, 50, 100
Tangential FM	-30, -10, 0, 10, 30	+/- 10, 20, 40, 80	10, 50, 100
4 CP-FSK	-30, -10, 0, 10, 30	10	10, 50, 100
Barker 5	-10, -30, 30, +20	N/A	10, 50, 100
Barker 7	-10, -30, 30, +20	N/A	10, 50, 100
Barker 11	-10, -30, 30, +20	N/A	10, 50, 100
Barker 13	-10, -30, 30, +20	N/A	10, 50, 100
Barker 5 $\otimes$ 11	-10, -30, 30, +20	N/A	10, 50, 100
Barker 5 $\otimes$ 13	-10, -30, 30, +20	N/A	10, 50, 100

For every waveform combination, the dataset contains a `NumPy` file with dimensions $[3, 1000, 33600]$, with dimensions [channels, pulses, samples]. The dimension definitions are summarised in Table 2.

Table 2. Structure of each waveform dataset file

Dimension	Size	Description
1	3	ADC channels
2	1000	Number of pulses acquired per waveform configuration
3	33600	I/Q samples per pulse (including noise padding)

The samples for each pulse are provided unprocessed, as digitised by the ADCs and formatted in the `np.int16` format with no additional synthetic noised added at this stage. To evaluate ES algorithms against a range of SNRs, a post-processing script is provided to add SNR variations into the data via adding varying levels of AWGN into the baseline samples.

The following scripts are provided as a starting template:

- **make_dataset.py** - A script that reads in each waveform file and then alters the SNR to a desired value range. This allows for the training and testing of machine learning models against a broad range of SNRs. The default range for this is -30 dB to +30 dB in 3 dB steps. From the raw dataset this script when ran "as-is" will create an augmented dataset, increased by a factor of 21.
- **plot_dataset.py** - A script that can be used to plot a Spectrogram plus the In-phase (I), Quadrature (Q), Magnitude, Phase (ϕ), Power (in dB) components of each pulse in the augmented dataset. By default, this script reads the first pulse for each waveform variation.
- **VGG13-Waveform-Classification-Example.ipynb** - A `jupyer notebook` that generates spectrogram training/testing data from the augmented dataset and then trains and evaluates a VGG13 based classifier for modulation recognition on a single ADCs data.

A. *Waveform Definitions*

The equations that describe the instantaneous frequency of the frequency modulated chirps are given below:

$$Linear\ f(t) = f_0 + (f_1 - f_0) * (\frac{t}{t_1}) \tag{1}$$

$$Exponential\ f(t) = f_0 + (f_1 - f_0) * (\frac{t}{t_1})^k \tag{2}$$

$$Quadratic\ f(t) = f_1 - (f_1 - f_0) * \frac{(t_1/2 - t)^2}{(t_1/2)^2} \tag{3}$$

The derivation of the instantaneous frequency of the tangential chirp is given in [2] by Widyantara et al.

$$Tan\ f(t) = \frac{f_1 + f_0}{2} - (f_0 - f_1) \cdot \frac{\tan\left(\frac{2\beta\,(t - \frac{t_1}{2})}{t_1}\right)}{2\,\tan(\beta)} \tag{4}$$

$$\text{where } \beta = \arctan(\alpha) \tag{5}$$

In half of the frequency-modulated examples, the start (f_0) and stop (f_1) frequencies were swapped to generate the corresponding "down" chirps.

For the continuous phase frequency shift keyed (CP-FSK) waveforms, four frequencies are chosen at random that satisfy the bandwidth and centre-frequency constraints provided. The pulse is split equally into sub-pulses which are each assigned one of the frequency values. To ensure spectral purity and minimise abrupt phase changes, the phase of each sub-pulse transitions continuously from the previous sub-pulse, maintaining phase continuity across frequency shifts.

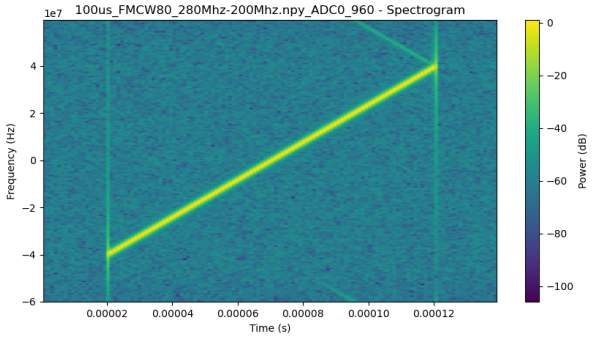


Fig. 1. Example Spectrogram from the dataset of a 80 MHz FMCW Chirp

In the binary Barker code implementation, the pulse is again split into sub-pulses, each of which is given a code value of "1" or "-1". Whenever the code changes between sub-pulses, the phase is shifted by 180 degrees.

Some compound barker codes are also present in the dataset. These are generated by taking the Kronecker product of two different barker sequences to create a longer code, which is split into sub-pulses and phase shifted using the same method described in the previous paragraph.

III. Hardware Description

The dataset introduced in this paper was transmitted and received over-the-air in the 2.4 GHz ISM (Industrial Scientific and Medical) band using the UCL "ARESTOR" system shown in Fig 2, which is an SDR (Software Defined Radio) based on an RFSoC (Radio Frequency System on Chip) architecture. ARESTOR has enabled experiments in active and passive radar [3], Integrated Sensing and Communications (ISAC) [4] [5] [6] as well as Electronic Surveillance (ES) [7] [8] [9]. More details about the ARESTOR system and it's hardware are provided in previous papers authored by the UCL Radar Group [10] [11].

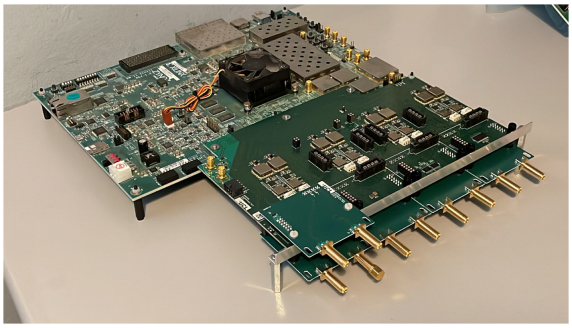


Fig. 2. The UCL ARESTOR System

To generate the RadarML dataset, a single ARESTOR node acting as both the transmitter and receiver(s) was used to avoid the requirement of synchronising multiple nodes. A single DAC (Digital-to-Analogue) channel was used as the transmitter

and 3 semi-independent ADC (Analogue-to-Digital) channels on different RF-Tiles were used for receive. Whilst the PLLs (Phase-Locked Loops) on each RF-Tile are locked to the same reference clock, no phase synchronisation (Multi-Tile Synchronisation) is performed between them, therefore the ADC channels are considered mesochronous.

The dataset waveforms were generated at an intermediate centre frequency of 240 MHz at a sample rate of 720 Ms/s before interpolation by a factor of 8 by the ADC tile on the RFSoC to achieve a final sample rate of 5.76 Gs/s. The waveforms are then digitally up-converted by mixing with a complex NCO (Numerically Controlled Oscillator) operating at a frequency of +2.2 GHz to enable transmission within the ISM band at a centre-frequency of 2.44 GHz.

The receivers on the RFSoC operate at a sample rate of at 3.84 Gs/s, which is then decimated in 32 stages (8 by the RF Tile and 4 by further FPGA processing) to achieve a final sample rate of 120 Ms/s. As the desired signal is sampled in the 2nd Nyquist zone, the actual intermediate frequency after sampling is given below in 6. The digitised signals are then digitally down converted using a NCO operating at -1.4 GHz, which mixes the centre-frequency of our target signals to baseband.

$$f_{\text{IF}} = |f_s - f_{\text{RF}}|, \quad \frac{f_s}{2} < f_{\text{RF}} < f_s \tag{6}$$

The receiver(s) are configured to record for 140 µs per acquisition, providing sufficient temporal coverage for all pulse durations in the dataset. Acquisitions are scheduled at 1 kHz to align with the transmitter's PRF of 1 kHz.

To ensure each transmitted pulse is centred within its corresponding receiver acquisition window, a variable pre-transmission delay is applied based on the pulse duration:

$$t_{\text{transmit delay}} = \frac{\left\lceil \left(1.4 \times 10^{-4} - t_{\text{pulse duration}}\right) \times 5 \times 10^{5} \right\rceil}{10^{6}}$$

Each receiver uses the same model of antenna (RFSPACE UWB3) and filter (Mini Circuits VBF-2435+), however, a different amplifier is used with each to introduce further variation into the received data. Attenuators are used to ensure each channel has roughly the same SNR regardless of the gain of the amplifier chosen. All the information on the physical hardware attached to each ADC channels used in this setup are described below in Table 3.

Table 3. DAC/ADC Channel Configurations

Port	Antenna	Amplifier	Filter	Attenuation
DAC0	L-COM HG2418DPD	MiniCircuits ZX60-6013E-S+	MiniCircuits VBF-2435+	N/A
ADC0	RFSPACE UWB3	MiniCircuits ZX60-6013E-S+	MiniCircuits VBF-2435+	N/A
ADC2	RFSPACE UWB3	MiniCircuits ZJL-4HG+	MiniCircuits VBF-2435+	1dB
ADC4	RFSPACE UWB3	MiniCircuits ZX60-83LN12+	MiniCircuits VBF-2435+	13dB

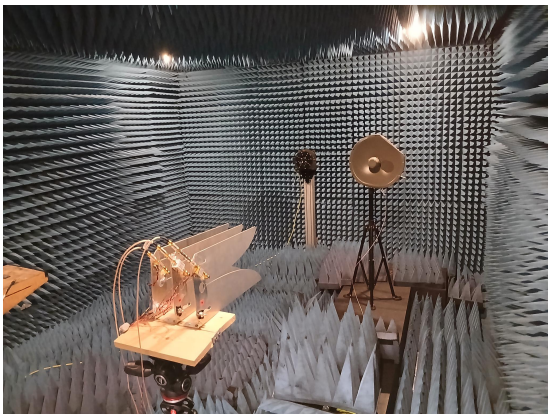


Fig. 3. Anechoic Chamber Setup

IV. Data and Code Availability

The dataset and some source code to get started are publicly available, the links for which are provided below:

- **Dataset:** The full dataset as described in this paper is available at https://doi.org/10.5522/04/30752767.
- **Example Processing Scripts:** The pre-processing and plotting scripts plus the example neural network as described in section II are available at https://github.com/UCLRadarGroup/Radar_ML.

The dataset is released under the CC BY 4.0 license and the code is released under the GPL-3.0 license.

V. Conclusions

This paper has presented RadarML, a experimentally captured multi-receive channel over-the-air dataset of representative radar waveforms designed to support the evaluation of electronic surveillance and modulation classification techniques. The data was generated using the UCL ARESTOR platform in order to systematically produce large quantities of a broad range of modulations and pulse parameter signals. ARESTOR operated as both the radar as well as the ES receiver in this case.

This dataset can now be utilised to benchmark a range of signal processing and Pulse Descriptor word (PDW) extraction methods. A key aspect is the multi-channel element as that allows for comparison of PDWs being extracted across all three channels, as well as the implementation of fusion methods to potentially improve the overall classification/estimation accuracies.

It is hoped that RadarML will enable reproducible comparison of algorithms and stimulate further research into multi-channel signal processing approaches.

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